

Uri Dickman

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EDUCATION

University of California-Santa Barbara

Santa Barbara, CA

Doctor of Philosophy, Mechanical Engineering

Sept. 2025 – Present

- GPA: 3.88
- *Fields:* Computational Science and Engineering; Solid Mechanics, Materials, and Structures

Brown University

Providence, RI

Bachelor of Science, Physics with Honors

Sept. 2020 – May 2024

- GPA: 3.9
- *Awards:* Undergraduate Teaching and Research Award (Summer 2023), Math Teaching Fellowship (Fall 2022)

RESEARCH EXPERIENCE

Graduate Student Researcher

September 2025 – Present

Computational Applied Science Laboratory, UC Santa Barbara

Santa Barbara, CA

- Developing hybrid neural operator-network architectures for learning solutions to free boundary problems. Applications include solidification of metal alloys and flows in porous media.

Research Assistant

July 2024 – August 2025

Computational Biomathematics Laboratory, Case Western Reserve University

Cleveland, OH

- Accomplishments: Developed a computational tool in Python for computation of complex models for Central Pattern Generator (CPG) circuits; dynamical systems analysis of a CPG to match experiments.
- Resulted in a first-author publication. Presented work at Society for Neuroscience Annual Meeting (2025).

Undergraduate Research Assistant

Jan. 2023 – May 2024

Solid State Nanofluidics Group, Brown University

Providence, RI

- Accomplishments: Detailed computational characterization of thermodynamics of water confined in carbon nanotubes. Began working on experimental apparatus. Included CVD, thermal evaporation and electrical characterization, Raman detection of CNTs. Resulted in 3rd-author publication.

WORK EXPERIENCE

Graduate Teaching Assistant

Sept. 2025 – Present

College of Engineering, UC Santa Barbara

Santa Barbara, CA

- Courses include: ME17 (Mathematics of Engineering), ME6 (Electronic and Electrical Circuits), and ME16 (Engineering Mechanics: Dynamics)
- Grading of homework and exams; holding office hours; leading lab sections.

Academic Tutor

July 2024 – May 2025

ModernSmart, Inc.

Remote

- Provide tutoring, exam preparation and curriculum creation for students in AP Physics and Calculus.

Undergraduate Teaching Assistant

Sept. 2021 – Aug. 2024

Depts. of Physics and Mathematics, Brown University

Providence, RI

- Teaching assistant for 9 courses, including Analytical Mechanics, Multivariable Calculus, and Calculus I.
- Grading of homework and exams; holding office hours; leading lab sections.

Instructional Laboratory Demonstrations Assistant

Feb. 2022 – May 2024

Dept. of Physics, Brown University

Providence, RI

- Designed and prepared live demonstrations for physics courses to convey physics concepts.

SKILLS & INTERESTS

Programming Languages: Python, C++, C, MATLAB, Mathematica, Julia

Scientific Tools: PyTorch, Linux, Slurm, MPI, NEURON, LAMMPS, L^AT_EX, Numpy, Zotero

Projects: Self-hosted Raspberry Pi high-performance computing cluster

Interests: Violin performance, chamber music, orchestra, digital music production